Yike Niu

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EDUCATION

Tsinghua University, Weiyang College 2024 – 2028 (Expected)
Mathematics and Physics + Mechanical Engineering GPA: 3.9/4.0
Selected coursework: Calculus (4.0), Linear Algebra (4.0), Data Structures and Algorithms (4.0), Mechatronic Measurement and Intelligent Control (4.0).

RESEARCH INTERESTS

- **Robot learning:** Efficient, reliable manipulation through visuomotor imitation learning.
- **Policy improvement:** Reinforcement learning, adaptation, and generalization through interaction.
- **Generalist robotics:** Vision-language-action models for complex, long-horizon manipulation.

PUBLICATIONS AND PREPRINTS

Jiale Cao*, **Yike Niu***, Zhengrong Xue, and Huazhe Xu. **AdaTempo: Learning Shared Relative Tempo from Demonstrations for Faster Robot Manipulation.**
Preprint, 2026. Submitted to ICRA 2027. arXiv:2610.02706 * *Equal contribution.*

Mingju Gao, Kaisen Yang, …, **Yike Niu**, …, Hao Zhao. **PAM: A Pose–Appearance–Motion Engine for Sim-to-Real HOI Video Generation.**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. Paper

RESEARCH EXPERIENCE

AdaTempo | Robot Learning and Manipulation Aug 2025 – Present
Advisor: Prof. Huazhe Xu, Tsinghua University

- **Method & Training**: Contributed to method design, demonstration processing, policy training, and hyperparameter tuning.
- **Simulation**: Led simulation experiments on robomimic and ALOHA, including ACT and Diffusion Policy comparisons and systematic ablation studies.
- **Real-World Evaluation**: Contributed to real-robot system setup, task design, data collection, and experimental evaluation across diverse manipulation tasks.
- **Outcome**: Co-authored an ICRA 2027 submission evaluating AdaTempo across 11 manipulation tasks, demonstrating up to 3.57× execution speedup over demonstrations.

PAM | Sim-to-Real Hand–Object Interaction Video Generation Sep – Dec 2025

- Processed and converted experimental data for sim-to-real hand–object interaction video generation.
- Contributed to qualitative analysis, result selection, and figure preparation for the CVPR 2026 paper.

SKILLS

Robot learning: ACT, Diffusion Policy, robomimic, ALOHA simulation tasks, policy training and evaluation.
Programming and ML: Python, C++, PyTorch; MATLAB (basic).
Platforms and tools: ALOHA, Franka, Linux, SolidWorks (CAD), LaTeX.

HONORS & AWARDS

Comprehensive Excellence Scholarship 2025 academic year
Academic Excellence Scholarship 2025 academic year
Third Prize — Weiyang City AI Agent Competition (Team Leader) May 2026
Second Prize — Chinese Mathematics Competitions Dec 2025
Third Prize — Regional University Physics Competition, Physics Division Dec 2025
Second Prize — Beijing College Student Mathematics Competition Nov 2025

LANGUAGES & INTERESTS

Languages: Chinese; English (TOEFL iBT 110/120).
Interests: Chinese cooking, baking, fitness, and basketball.